GEOSPATIAL COMPUTER VISION & EARTH OBSERVATION DATA AUDIT

Electro-Optical (EO) Data Audit & Exploratory Data Analysis: DisasterM3 vs. BRIGHT

Strict Optical-Only Comparative Assessment of Pre-Event EO Baselines in BRIGHT (ESSD 2025) and Bi-Temporal Pre/Post-Event EO Pairs in DisasterM3 (NeurIPS 2025) — Excluding All SAR Modalities

4,246
BRIGHT Pre-Event EO Tiles (0.3–1.0m GSD)

22,214
DisasterM3 Total EO Images (0.8m GSD)

11,107
DisasterM3 Pure EO Bi-Temporal Pairs

384,596
BRIGHT Audited Building Footprints

1. Executive Audit Scope & Architectural Divergence

SECTION 01 // SCOPE

This data audit and Exploratory Data Analysis (EDA) conducts a rigorous, strictly Electro-Optical (EO) comparison between two flagship remote sensing disaster benchmarks developed by overlapping research teams at The University of Tokyo, RIKEN AIP, and ETH Zürich: **BRIGHT** (Chen et al., 2025) and **DisasterM3** (Wang et al., 2025). In accordance with audit parameters, all Synthetic Aperture Radar (SAR) imagery (Capella Space and Umbra VV/HH amplitude channels) is systematically excluded from the primary evaluation to isolate the optical sensor characteristics, spatial resolution trade-offs, radiometric pipelines, and bi-temporal optical pairing mechanics.

While both datasets address global disaster damage assessment, their optical architectures serve fundamentally different operational paradigms. **BRIGHT** treats VHR optical imagery strictly as a *pre-event structural baseline* (T_1) across 4,246 tiles (1024×1024 pixels) spanning 14 events, relying on optical RGB channels to delineate crisp building footprints and pre-disaster roof geometries—except in a specialized 5-event controlled subset where post-event optical data (T_2) is introduced to benchmark pure Optical-Optical upper bounds. Conversely, **DisasterM3** expands the scope into a multi-task Vision-Language Model (VLM) benchmark across 36 events. Out of its 26,988 total images (forming 15,881 multi-temporal pairs), exactly **22,214 are optical images** and 4,774 are post-event SAR images. Deductions from the pairing architecture reveal that DisasterM3 provides **11,107 pure Optical-Optical bi-temporal pairs** (22,214 − 4,774 = 17,440 optical images in pure EO pairs, plus 4,774 pre-event optical baselines paired with SAR).

Audit Dimension	BRIGHT Dataset (Optical Component Only)	DisasterM3 Dataset (Optical Component Only)
Primary EO Role	**Pre-Event Baseline (T1)** Unimodal optical localization anchor; 5-event T_1+T_2 EO ablation subset.	**Bi-Temporal EO Pairs (T1+T2)** 11,107 pure Optical-Optical pairs + 4,774 Pre-EO baselines.
Total Optical Volume	**4,246 pre-event EO tiles** (1024×1024 px) across 14 historical disaster events (23 study regions).	**22,214 optical images** (17,190 in *Instruct* split; 5,024 in *Bench* split) across 36 events.
Ground Sampling Distance	**Native Variable VHR: 0.30 m to 1.00 m/pixel** (preserves raw sensor fidelity per region).	**Harmonized Fixed: 0.80 m/pixel** (all pre- and post-disaster optical scenes resampled).
Spectral & Radiometric	RGB (Red, Green, Blue); Digital Numbers (DN) converted to surface reflectance, standardized to 8-bit [0, 255].	RGB optical channels; WorldView/Maxar archive standardized and georeferenced to 0.8 m grid.
EO Sensor Provenance	Maxar Open Data (WorldView), Google Earth, NOAA, NAIP (USA), IGN (Spain), GSI (Japan).	xBD optical archive (15 events), BRIGHT optical archive (11 events), + 10 new Maxar Open Data events.
Geographic & Event Scope	14 events (2020–2024): 5 natural & 2 human-made hazard classes; focused on developing nations.	36 events across 5 continents: 10 natural & human-made hazard classes (6,750 fire; 5,279 EQ; 4,834 hurricane).
Annotated Entities	**384,596 building polygons** (Pixel-level: Background 0, Intact 1, Damaged 2, Destroyed 3).	Instance masks for **Buildings** (Intact, Damaged, Destroyed) & **Roads** (Intact, Flooded, Debris-covered) + 123k VLM QA pairs.

2. Complete Event-Level Audit of BRIGHT Pre-Event EO Baselines SECTION 02 // BRIGHT EO

Unlike DisasterM3, which enforces a uniform 0.80 m/pixel resampling across all optical scenes, BRIGHT retains the native spatial resolution of its heterogeneous Electro-Optical providers (0.30 m to 1.00 m GSD). The table below audits all 14 disaster events in BRIGHT, detailing the optical sensor provenance, exact GSD, tile counts, building footprints, proxy registration errors (RMSE in pixels between optical baselines and post-event grids), and inclusion in the pure post-event EO benchmark subset.

Disaster Event & Code	Date	EO Provider	EO GSD (m)	EO Tiles	Buildings	Reg. RMSE (All / Bldg)	Post-EO Subset?
Beirut, Lebanon (EP)	04 Aug 2020	Maxar Open Data	1.00 m	133	25,496	1.365 / 1.365 px	**Yes (T1+T2 EO)**
Bata, Eq. Guinea (EP)	07 Mar 2021	Maxar Open Data	0.50 m	107	8,893	1.226 / 1.160 px	**Yes (T1+T2 EO)**
Goma, DR Congo (VE)	22 May 2021	Maxar Open Data	0.33 m	123	18,741	0.966 / 0.975 px	Pre-EO Only
Les Cayes, Haiti (EQ)	14 Aug 2021	Maxar Open Data	0.48 m	73	18,918	1.392 / 1.377 px	Pre-EO Only
La Palma, Spain (VE)	19 Sep 2021	IGN (Spain)	0.30–0.35 m	933	30,239	0.976 / 0.944 px	Pre-EO Only
Boulder (Marshall), USA (WF)	30 Dec 2021	NAIP Aerial EO	0.60 m	77	8,365	1.263 / 1.266 px	Pre-EO Only
Ukraine Conflict (AC)	Mar–Sep 2022	Google Earth EO	0.60 m	513	56,770	0.899 / 0.882 px	Pre-EO Only
SE Turkey (EQ)	06 Feb 2023	Maxar Open Data	0.30–0.35 m	1,114	135,033	0.937 / 0.901 px	Pre-EO Only
Kyaukpyu, Myanmar (CC)	14 May 2023	Google Earth EO	0.60 m	126	8,052	0.970 / 0.998 px	Pre-EO Only
Maui, Hawaii, USA (WF)	08 Aug 2023	NOAA Aerial EO	0.60 m	65	3,995	1.124 / 1.124 px	**Yes (T1+T2 EO)**
Al Haouz, Morocco (EQ)	08 Sep 2023	Maxar Open Data	0.35–0.40 m	567	6,269	1.088 / 1.050 px	Pre-EO Only
Derna, Libya (FL)	10 Sep 2023	Maxar Open Data	0.35 m	124	10,979	1.267 / 1.255 px	**Yes (T1+T2 EO)**
Acapulco, Mexico (HC)	25 Oct 2023	Google Earth EO	0.35–0.80 m	212	18,437	1.166 / 1.148 px	Pre-EO Only
Noto Peninsula, Japan (EQ)	01 Jan 2024	GSI (Japan) EO	0.50 m	79	8,153	1.072 / 1.043 px	**Yes (T1+T2 EO)**
BRIGHT Total / Aggregate	**2020–2024**	**6 EO Sources**	**0.30–1.00 m**	**4,246**	**384,596***	**1.024 / 1.006 px**	**5 Events (488 Tiles)**

3. Genealogical Lineage, Resolution Loss & Radiometric Audit SECTION 03 // LINEAGE & GSD

A critical discovery of this data audit is the direct genealogical inheritance between BRIGHT and DisasterM3, alongside a subtle arithmetic anomaly in BRIGHT's published tabular metadata:

- **Dataset Inheritance & Resampling Degradation (0.3m → 0.8m):** DisasterM3 constructs its 36-event archive by absorbing **26 events from xBD (15 events) and BRIGHT (11 events)**, plus **10 newly curated events** from Maxar's Open Data program (including Hurricane Ian, Shovi Landslide, Kalehe Landslide, and Nepal Flooding). Crucially, when BRIGHT's sub-meter optical tiles—such as *Turkey-EQ* (0.30–0.35 m), *La Palma-VE* (0.30–0.35 m), *Goma-VE* (0.33 m), and *Derna-FL* (0.35 m)—are ingested into DisasterM3, they are downsampled to a uniform **0.80 m/pixel GSD**. This represents a **2.28× to 2.67× linear resolution reduction** (and a **5.2× to 7.1× loss in pixel area density**), smoothing out

4. Exploratory Data Analysis: Distributions & Class Imbalance

SECTION 04 // STATISTICAL EDA

Both datasets exhibit extreme class imbalance and pronounced hazard-type skew within their optical annotations. In BRIGHT, non-building background pixels dominate at **85.9%** vs. **14.1%** building foreground pixels (an approximate 6.1:1 background-to-foreground ratio, reported as ~7:1). Within building pixels/instances, undamaged structures overwhelmingly dominate both datasets, creating severe long-tail optimization challenges for minority damage tiers.

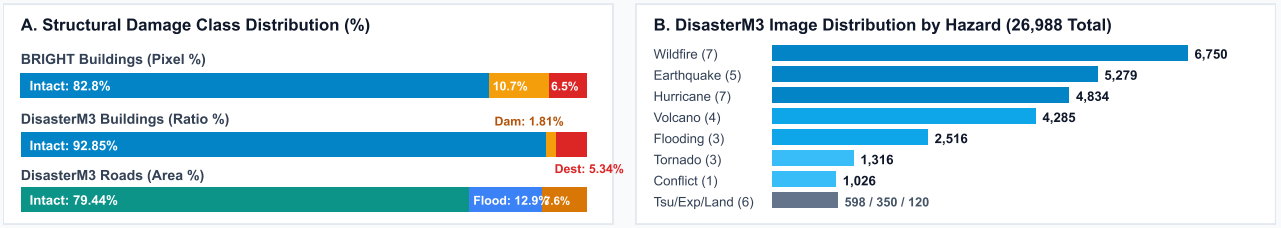


Figure 1: (A) Comparison of building and road damage class distributions across BRIGHT and DisasterM3. (B) Hazard-type image distribution in DisasterM3 (note: sum of individual bars is 27,074 vs. 26,988 stated in text due to 86 multi-hazard overlapping tiles).

A comparative audit of the class distributions reveals a striking structural divergence in intermediate damage representation. In BRIGHT, **Intact** buildings account for **82.8%**, **Damaged** for **10.7%**, and **Destroyed** for **6.5%**. In DisasterM3, because of the massive influx of wide-area xBD wildfire and hurricane tiles with vast unaffected suburban perimeters, the **Intact** building ratio rises to **92.85%**, while **Destroyed** is **5.34%** and **Damaged** collapses to just **1.81%**. This 5.9× reduction in the relative proportion of partially damaged buildings in DisasterM3 makes fine-grained roof-damage classification substantially harder, whereas BRIGHT's manual disaggregation of block-level UNOSAT/Copernicus polygons preserved a much richer 10.7% intermediate damage tier. However, DisasterM3 compensates by introducing pixel-level **Road Damage masks** across three actionable categories: **Intact (79.44%)**, **Flooded / Water-Blocked (12.94%)**, and **Debris-Covered (7.62%)**.

5. Deep-Dive Audit: Pre-Event Baseline vs. Bi-Temporal EO Pairs

SECTION 05 // OPTICAL ROLES

A. Pre-Event EO Baseline in BRIGHT (T_1 Role)

In BRIGHT, the pre-event optical image X_{T1} is far more than a passive geometric template for building localization. Controlled ablation experiments in Section 4.3 (Table 7)—where ground-truth building footprint masks were supplied to isolate localization from classification—prove that adding pre-event optical imagery to post-event inputs boosts **UNet** damage classification F_1^{clf} from **68.71% to 73.59% (+4.88%)** and surges the "Damaged" class IoU from **35.83% to 44.83% (+9.00%)**. Similarly, **DeepLabV3+** sees its "Destroyed" class IoU jump from **59.54% to 64.94% (+5.40%)**.

Grad-CAM visualizations (BRIGHT Figure 7) and DINOv2 t-SNE embeddings (Figure 5c) confirm that pre-event VHR optical imagery encodes critical **pre-disaster vulnerability and structural semantics**: roof pitch, construction material reflectance, building scale (which follows a heavy long-tail distribution up to 12,000 pixels per structure), and surrounding

B. Pre/Post-Event EO Pairs in DisasterM3 (T_1+T_2 Role)

DisasterM3 operationalizes bi-temporal Optical-Optical pairs across 9 perception and reasoning tasks spanning 5 core capabilities: **Disaster Recognition** (Disaster Type DTR, Scene Land-Use DSR across 13 AID classes, Bearing-Body BBR across 12 infrastructure classes), **Damage Counting** (Building Counting BDC, Road Area Estimation DRE), **Pixel Grounding** (Referring Segmentation of damaged structures, lava proximity buffers within 100m, helicopter staging track fields, and dispatch vehicle parking lots), **Relational Reasoning** (ORR across bounding-box pairs), and **Long-Form Reporting** (Disaster Caption & Immediate/Long-Term Restoration Advice).

In DisasterM3's Optical-Optical benchmark, pre-event EO establishes the baseline land-use taxonomy (e.g., port, industrial, residential, viaduct) and intact object inventory, while post-event EO provides high-frequency chrominance and texture changes;

6. Benchmark Performance Audit on Pure Electro-Optical Pairs

SECTION 06 // EO BENCHMARKS

To evaluate how modern AI architectures exploit pre- and post-event optical data, the table below synthesizes benchmark metrics strictly from the **pure Electro-Optical settings** of both papers: Part I audits pixel-level semantic segmentation networks on BRIGHT's 5-event Optical-Optical subset (Table 8), while Part II audits Vision-Language Models on DisasterM3's Optical-Optical Bench split (Tables 2, 3, and 9).

Dataset & Architecture	Model Paradigm	Loc / QA Metric	Clf / DTR / BBR	Counting (BDC / RMSE)	Primary Segmentation / Report Score
BRIGHT (EO-EO): UNet	Coupled CNN Seg.	$F_1^{loc.}$: 86.46%	$F_1^{clf.}$: 75.64%	Intact IoU: 68.62%	**mIoU: 65.96%** (Dam: 45.27% / Dest: 55.36%)
BRIGHT (EO-EO): DeepLabV3+	Coupled ResNet-50	$F_1^{loc.}$: 85.79%	$F_1^{clf.}$: 74.39%	Intact IoU: 67.90%	**mIoU: 64.87%** (Dam: 44.31% / Dest: 52.94%)
BRIGHT (EO-EO): DamageFormer	Decoupled Swin-T	$F_1^{loc.}$: **88.32%**	$F_1^{clf.}$: **78.04%**	Intact IoU: **72.72%**	**mIoU: 69.76%** (Dam: **47.26%** / Dest: **63.68%**)
DisasterM3 (EO-EO): TeoChat-7B	Temporal RS VLM	QA AVG: 23.0%	64.9% / 2.0%	BDC: 22.5% (DRE: 23.3%)	Caption: 1.77 / 5.0 | Advice: 1.95 / 5.0
DisasterM3 (EO-EO): EarthDial-4B	Multi-Sensory RS VLM	QA AVG: 22.9%	58.1% / 3.2%	BDC: 30.2% (DRE: 20.8%)	Caption: 1.53 / 5.0 | Advice: 2.42 / 5.0
DisasterM3 (EO-EO): Qwen2.5-VL-7B	Open VLM (Zero-Shot)	QA AVG: 31.2%	66.6% / 4.7%	BDC: 34.2% (RMSE: 69.66)	Caption: 1.75 / 5.0 | Advice: 1.95 / 5.0
DisasterM3 (EO-EO): Qwen2.5-VL-72B	Open VLM (Scaled)	QA AVG: 40.5%	74.8% / 6.8%	BDC: **34.8%** (RMSE: **53.83**)	Caption: 2.01 / 5.0 | Advice: 2.92 / 5.0
DisasterM3 (EO-EO): InternVL3-8B	Open VLM (Zero-Shot)	QA AVG: 31.3%	53.5% / 4.0%	BDC: 30.3% (RMSE: 86.93)	Caption: 1.96 / 5.0 | Advice: 2.75 / 5.0
DisasterM3 (EO-EO): InternVL3-78B	Open VLM (Scaled)	QA AVG: 39.3%	72.5% / 5.3%	BDC: 29.4% (RMSE: 105.96)	Caption: 2.79 / 5.0 | Advice: 2.90 / 5.0
DisasterM3 (EO-EO): GPT-4.1	Commercial VLM	QA AVG: **42.3%**	79.6% / 7.2%	BDC: 25.5% (RMSE: 115.89)	Caption: 2.57 / 5.0 | Advice: 3.14 / 5.0 (ORR: **64.0%**)
DisasterM3 (EO-EO): Qwen2.5-VL-7B-FT	LoRA Fine-Tuned VLM	QA AVG: 40.4%	**83.6%** / 21.5%	BDC: 34.3% (RMSE: 61.39)	Caption: **3.90 / 5.0** | Advice: 3.11 / 5.0
DisasterM3 (EO-EO): InternVL3-8B-FT	LoRA Fine-Tuned VLM	QA AVG: **41.7%**	79.3% / **23.9%**	BDC: 29.1% (RMSE: 108.88)	Caption: 3.83 / 5.0 | Advice: **3.31 / 5.0**
DisasterM3 (EO-EO): LISA-7B (ZS → FT)	Referring Seg. VLM	cIoU: 22.1→43.7%	Road: 11.9→27.6%	Bldg: 25.0% → 41.2%	**mIoU: 27.5% → 44.8% (+17.3%)** | Other: 65.5%
DisasterM3 (EO-EO): PSALM-1.3B (ZS → FT)	Mask2Former VLM	cIoU: 6.3→**44.5%**	Road: 2.6→**30.5%**	Bldg: 10.2% → **49.1%**	**mIoU: 9.7% → 50.5% (+40.8%)** | Other: **71.9%**

7. Critical EDA Insights, Failure Modes & Recommendations

SECTION 07 // SYNTHESIS

- **Task Decoupling vs. Multi-Token Query Decoding in EO Segmentation:** In BRIGHT's optical benchmark, decoupling building localization (T_1 EO) from damage classification (T_1+T_2 EO) via *DamageFormer* outperforms coupled *UNet* by **+3.80% mIoU** and **+8.32% Destroyed IoU**. Similarly, in DisasterM3's Optical-Optical referring segmentation, *PSALM-1.3B* (which uses 100 learnable Mask2Former query tokens) outperforms *LISA-7B* (which uses a single fixed mask token) by **+5.7% mIoU (50.5% vs. 44.8%)** and **+7.9% Building IoU (49.1% vs. 41.2%)** despite having 5.4× fewer parameters. Multi-scale optical debris and building clusters require multi-query token representations rather than single-bottleneck embeddings.
- **Scene Complexity & Density-Induced Counting Overfitting in Optical Pairs:** Across DisasterM3's 10 hazard categories